Alapan Das

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Professional Summary

Final-year Information Technology student with strong skills in programming, systems management and problem solving. Known for adaptability, attention to detail, time management, and effective team collaboration. Eager to apply academic learning to real-world IT challenges and contribute to impactful solutions.

Technical Skills

Programming Languages: C++, Core Java, C, Python,Javascript
Database: MySQL, Oracle
Web Technologies: HTML, CSS, Node.js
Version Control: Git, Github
Development Methodologies: Agile, Scrum, Kanban, Prototype, Spiral
Deployment Tools: Docker, Vercel

Experience

IBM SkillsBuild Internship *June-Aug'24*

- Completed a Front-End Web Development task for **FoodWise** using **HTML**, **CSS**, and **JavaScript**.
- Collaborated in a team environment to design user interfaces and resolve integration issues with RESTful APIs.
- Gained exposure to Agile practices and version control using **GitHub**.

Projects

Exploring the Potential of De Bruijn Graph Topology to Serve Next-Generation Data Centres (Group Project) **(Documentation)**

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Leaf–Spine vs De Bruijn Topology Study

- Designed and analyzed scalable data center network architectures, comparing traditional Leaf–Spine with graph-based De Bruijn topology.
- Studied hop-count growth, routing determinism, and latency behavior, showing reduced routing overhead in De Bruijn networks.
- Investigated load balancing techniques (ECMP, packet spraying) and their impact on TCP performance.

AI Powered Smart Traffic Management System (Group Project) **(Github Repository)**

Flask, PyTorch, OpenCV, Flask-SocketIO, NetworkX, RESTful APIs

- Developed a real-time smart traffic light system using **DQN**, **PyTorch**, and **OpenCV** for video-based traffic density classification, enabling dynamic signal timing and congestion reduction through **NetworkX**-based alternate route engine and multi-intersection synchronization.
- Built **RESTful APIs**, **WebSocket** endpoints, and integrated **Flask-SocketIO** for live updates, route suggestions, real-time video streaming, and seamless backend-frontend interaction via interactive dashboards.

Driver's Drowsiness Alert System (Group Project) **(Github Repository)**

Python, Flask, OpenCV, Dlib, eSpeak, NumPy, Imutils, HTML, CSS

- Developed a real-time driver drowsiness detection system using **OpenCV**, **Dlib**, and **Flask** for live facial monitoring, implementing **facial landmark detection** and eye movement tracking to identify fatigue and trigger alerts.
- Designed a web-based interface for accessibility and integrated multi-threaded voice alerts using Python's threading module and **eSpeak** for real-time accident prevention.

Achievements

AI Powered Smart Traffic Management System *May'25*

•Finalist in Smart Bengal Hackathon

Publications

Exploring the Potential of De Bruijn Graph Topology to Serve Next-Generation Data Centres *Mar'26*

•Accepted at the 13th International Symposium on Applied Computing for Software and Smart Systems

Education

B.Tech in **Information Technology** from *University of Calcutta* (**CGPA:** 8.15) *2023-2026(Expected)*
B.Sc in **Computer Science (Hons.)** from *University of Calcutta* (**CGPA:** 8.379) *2020-2023*
Higher Secondary Education (*W.B.C.H.S.E*)(**Percentage:** 88.4%) *2020*
Secondary Education (*W.B.B.S.E*)(**Percentage:** 87.4%) *2018*

Soft Skills

Problem-solving, Team-Collaboration, Time Management, Attention to Detail, Adaptability